

Medium Access Protocol, Internet Protocol and Transport Layer

1. Medium Access Protocol (MAC)

Definition

A **Medium Access Protocol (MAC)** is a set of rules that controls **how multiple devices access and share a communication channel in a network**. It ensures that devices transmit data efficiently without collisions. ([GeeksforGeeks](#))

Exam Definition

Medium Access Protocol is a protocol that **regulates how multiple devices share and access a common communication medium in a network**.

Need for Medium Access Protocol

When many devices try to send data over the **same channel**, collisions can occur. MAC protocols help by controlling **when and how devices transmit data** to avoid interference. ([GeeksforGeeks](#))

Types of Medium Access Protocols

Type	Explanation
Random Access Protocols	Devices transmit when the channel is free
Controlled Access Protocols	Devices transmit in a controlled sequence
Channelization Protocols	Channel divided into multiple parts for users

Examples

- ALOHA
 - Slotted ALOHA
 - CSMA/CD
 - CSMA/CA
 - TDMA
 - FDMA
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Advantages

- Reduces collisions
 - Improves network efficiency
 - Ensures fair communication between devices
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2. Internet Protocol (IP)

Definition

Internet Protocol (IP) is a **network layer protocol** used for **addressing and routing packets across networks** so that data reaches the correct destination.

([GeeksforGeeks](#))

Exam Definition

Internet Protocol is a protocol that **assigns logical addresses (IP addresses) to devices and routes packets between networks**.

Functions of Internet Protocol

- **Logical Addressing** – Assigns unique IP addresses to devices
 - **Routing** – Determines the path for data packets
 - **Packet Fragmentation** – Divides large packets into smaller ones
 - **Packet Delivery** – Sends packets from source to destination network
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Types of IP

Protocol	Description
IPv4	Uses 32-bit addressing
IPv6	Uses 128-bit addressing

Advantages

- Enables **communication between different networks**
 - Supports **global Internet connectivity**
 - Provides **logical addressing system**
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3. Transport Layer

Definition

The **Transport Layer** is responsible for **end-to-end communication between devices** and ensures reliable data transfer between applications. ([GeeksforGeeks](#))

Exam Definition

Transport Layer is the **fourth layer of the OSI model** that provides **process-to-process communication and reliable data delivery between hosts**. ([GeeksforGeeks](#))

Functions of Transport Layer

- **End-to-End Communication** between devices
- **Segmentation** – Divides data into smaller segments
- **Flow Control** – Controls transmission speed
- **Error Control** – Detects and corrects errors
- **Multiplexing/Demultiplexing** using port numbers

These features help ensure **reliable and efficient communication between applications**. ([GeeksforGeeks](#))

Transport Layer Protocols

Protocol	Description
TCP (Transmission Control Protocol)	Reliable and connection-oriented communication
UDP (User Datagram Protocol)	Fast but connectionless communication

TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable	Less reliable
Speed	Slower	Faster
Example Use	Email, Web browsing	Streaming, gaming
